

Recovering a Bricked Dell PowerEdge R730

Flashing BIOS When the Server Won't POST

Escaping the BIOS 1.0.4 + Lifecycle Controller 2.02 dead-end
on the R730 (13G) using only a *basic* iDRAC license

A searchable field guide for anyone hitting the same wall

TL;DR

If you put modern high-density DIMMs (e.g. 32 GB LRDIMMs) into an R730 that still has its **2014 factory BIOS (1.0.4)** and **Lifecycle Controller 2.02.01.01**, the server fails to POST with **No useable DIMMs found**, and you cannot update the BIOS because:

- The machine never reaches the **F11 / F2 boot menu** (BIOS 1.0.4 faults on USB init first).
- The **iDRAC web UI** speaks only TLS 1.0/1.1, which every modern browser refuses.
- **racadm update** over SSH is blocked — you have only a **Basic/Express** license, not Enterprise.
- **racadm fwupdate** (the legacy TFTP path) rejects a BIOS **.exe** — it accepts only an iDRAC **.d7**.
- **Lifecycle Controller 2.02** itself is **broken** for local/network updates (HWC0005).

The escape

Update the **iDRAC/Lifecycle Controller** first via the legacy **fwupdate** TFTP method (which *does* accept the iDRAC **.d7**), jumping it to **2.86.86.86**. That fixes the browser TLS problem. Then use the now-working **iDRAC 2.86 web UI** to upload and flash the BIOS **.exe** directly — the web-UI upload path does **not** require Enterprise. BIOS goes **1.0.4** → **2.19.0**, the high-density RAM trains, and the server POSTs.

Searchable key phrases: *R730 won't POST after RAM upgrade; No useable DIMMs found; BIOS 1.0.4 won't update; Lifecycle Controller 2.02 HWC0005; iDRAC TLS 1.0 browser; flash R730 BIOS without Enterprise license.*

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1 The Hardware

Component	Detail
Server	Dell PowerEdge R730 (13th Generation / “13G”)
CPUs	Dual Intel Xeon E5-26xx v4 (Broadwell-EP)
Target RAM	512 GB = 16 × 32 GB DDR4-2133 LRDIMM (Samsung M386A4G40DM0-CPB)
Starting BIOS	1.0.4 (factory, dated 08/28/2014)
Starting iDRAC	iDRAC8, firmware 2.02.01.01
Starting LC	Lifecycle Controller 2.02.01.01
iDRAC license	Basic / Express (NOT Enterprise)
System ID	Decimal 1536 = hex 0x0600

Component	Detail
Helper host	A second Linux box (Debian 12) on the same management subnet

Terminology

The **helper host** is any second machine on the same L2 network as the iDRAC that can run `ssh`, `tftpd-hpa`, and a browser. In the original session it was a Debian 12 workstation at 192.168.10.199; the iDRAC was at 192.168.10.20. Substitute your own IPs throughout.

2 The Symptom

After swapping the original lower-density DIMMs for the 32 GB LRDIMMs, the server would **not POST**. On screen: **No useable DIMMs found**. The iDRAC System Event Log (SEL) showed, repeatedly: **No memory detected**.

Important *misleading* observations:

- **No amber/fault LEDs** anywhere on the board or at any DIMM slot.
- The hang happened **regardless** of DIMM count (1, 2, or 8 sticks), slot, or CPU bank.
- It **even happened with known-good lower-density DIMMs** from a second, working R730.

That last point sends most people down the wrong path (suspecting dead hardware). It is not dead hardware.

3 Root Cause

BIOS 1.0.4 cannot train the memory and, combined with Lifecycle Controller 2.02, leaves no working update path. The factory BIOS shipped in 2014. The failure is in the earliest memory-init/training phase, so the machine powers down (or loops) *before* you can reach any menu, boot device, or recovery tool.

Two community contributors independently confirmed that **Xeon v4 CPU support was a red herring** — that support predates this BIOS — correctly pointing at the **BIOS image + broken LC**, not the CPUs or RAM.

4 The Catch-22

Can't POST, because BIOS 1.0.4 can't train the RAM. Can't update the BIOS, because every update path needs something you can't reach without POST: a boot menu, a modern-TLS browser, an Enterprise license, or a working Lifecycle Controller.

You break the loop by realizing the **iDRAC out-of-band processor is alive the whole time** — it runs on standby power, independent of POST — and that the **legacy fwupdate TFTP path can update the iDRAC/LC** even though it can't update the BIOS. Update the iDRAC/LC first; that restores a usable web UI; the web UI then flashes the BIOS.

5 Things That Did NOT Work (and why)

Every one of these was attempted, in order. Knowing what to skip saves hours.

5.1 RAM / hardware permutations (all failed identically)

Attempt	Result
Remove one CPU's bank, leave 8×32 GB on the other	No useable DIMMs found
Drop to 2 sticks in A1 + B1	Same
Drop to a single stick in A1 (multiple sticks tried)	Same
Install known-good lower-density DIMMs from a 2nd R730	Same
Move those DIMMs to the <i>other</i> CPU's slots	Same
Reseat everything, clean contacts	Same
Swap CPU1 for a known-good Xeon E5-2650 v4	Same
Inspect CPU1 socket under magnification	No bent pins
Clear CMOS via PWRD_EN / NVRAM_CLR jumpers	No change

Takeaway

If your R730 fails to detect memory *no matter what you put in it* **and there are no amber LEDs**, stop swapping hardware. Suspect firmware.

5.2 BIOS-update paths that are dead-ends on 1.0.4 + LC 2.02

Path	Why it failed
Bootable USB with BIOS .exe	BIOS 1.0.4 faults during USB enumeration; F11/F2 never reachable with the memory error present.
Internal motherboard USB	Same — no auto-flash triggered.
iDRAC web UI (on 2.02)	iDRAC 2.02 serves only TLS 1.0/1.1; Chrome and Firefox both refuse. Chrome's thisisunsafe does not bypass a hard TLS-version mismatch.
racadm update over SSH	SWC0242: required license missing — needs Enterprise.
racadm fwupdate + BIOS .exe	Wrong artifact. fwupdate accepts only an iDRAC .d7.
LC GUI update (on 2.02)	HWC0005 — LC 2.02 has a known update bug.
One-time boot to LC	Server dies on the memory error before LC loads.
Hand-built Dell catalog (NFS/-CIFS/FTP/HTTP)	SWC0053 / SWC0066 / SUP0530 / SUP0539; Dell's current catalog no longer lists the R730.
Enterprise <i>trial</i> license (3rd party)	LIC017 — trial licenses are device-locked to a specific Service Tag.

6 Solution, Stage 1 — Update iDRAC/LC via TFTP

Goal: get iDRAC + Lifecycle Controller from 2.02 → **2.86.86.86**, which fixes the browser TLS problem and gives a working LC. Use the legacy **fwupdate** TFTP path, because it accepts an iDRAC

.d7 and does not require Enterprise.

6.1 Confirm the iDRAC is reachable

The iDRAC runs on standby power and is alive even though the host won't POST. Find its IP (the R730 front-panel LCD shows it), then from the helper host:

```
ssh root@<IDRAC_IP>          # default password: calvin
racadm getversion             # confirm BIOS 1.0.4 / iDRAC 2.02 / LC 2.02
```

6.2 Download the Linux .BIN (not the Windows .exe)

From Dell's support site for your Service Tag, under **iDRAC with Lifecycle Controller**, download the Linux payload:

```
iDRAC-with-Lifecycle-Controller_Firmware_VWF72_LN64_2.86.86.86_A00.BIN
```

The LN64 .BIN is a self-extracting Linux package. The Windows .exe is useless here because the host has no OS.

6.3 Extract the .d7 firmware out of the .BIN

```
sudo /path/to/iDRAC-..._LN64_2.86.86.86_A00.BIN --extract /tmp/idrac_extract
find /tmp/idrac_extract -name 'firmimg.d7'
# -> /tmp/idrac_extract/payload/firmimg.d7
```

`--extract` must be used *alone* (it can't be combined with other flags) and needs `root`.

6.4 Set up TFTP — and mind the exact path iDRAC asks for

Debian's `tftpd-hpa` serves from `/srv/tftp` (not `/var/lib/tftpboot`), and may ship **masked**:

```
sudo apt-get install -y tftpd-hpa
sudo systemctl unmask tftpd-hpa
sudo systemctl start tftpd-hpa
systemctl status tftpd-hpa      # confirm it serves /srv/tftp
```

When you point `fwupdate` at a file called `idrac.bin`, the iDRAC actually requests `idrac.bin/firmimg.d7` over TFTP — it treats `idrac.bin` as a *directory*. Lay the files out to match:

```
sudo rm -f /srv/tftp/idrac.bin
sudo mkdir -p /srv/tftp/idrac.bin
sudo cp /tmp/idrac_extract/payload/firmimg.d7 /srv/tftp/idrac.bin/firmimg.d7
```

The biggest gotcha

If `firmimg.d7` is not at `/srv/tftp/idrac.bin/firmimg.d7`, `fwupdate` returns `ERROR: Remote host is not reachable or connection is interrupted` — which looks like a network problem but is a **file-not-found** at the requested path. Confirm with:

```
sudo tcpdump -i any -n udp port 69
# -> RRQ "idrac.bin/firmimg.d7"
```

6.5 Open the firewall for TFTP (UDP/69)

```
sudo iptables -I INPUT -p udp --dport 69 -s <IDRAC_IP> -j ACCEPT
sudo iptables -I INPUT -p udp --sport 69 -s <IDRAC_IP> -j ACCEPT
```

(ufwequivalent: `sudo ufw allow from <IDRAC_IP> to any port 69 proto udp`.) Optionally verify the file is served:

```
tftp <HELPER_HOST_IP> -c get idrac.bin/firmimg.d7 /tmp/test.d7
ls -la /tmp/test.d7      # ~115 MB
```

6.6 Fire the iDRAC/LC update

```
ssh root@<IDRAC_IP> "racadm fwupdate -g -u -a <HELPER_HOST_IP> -d idrac.bin"
```

Expect:

```
Preparing for firmware update. Please wait...
Firmware update completed successfully. The RAC is in the process of
resetting. Your connection will be lost. Please wait for a few minutes
before starting a new session.
```

Wait 3–5 minutes, then confirm:

```
ssh root@<IDRAC_IP> "racadm getversion"
# iDRAC Version = 2.86.86.86
# Lifecycle Controller Version = 2.86.86.86
# BIOS Version = 1.0.4 (still -- that's next)
```

racadm filename limit

`racadm` rejects very long filenames (`ERROR: Specified path is too long`). Use the short name `idrac.bin` instead of the full Dell filename.

7 Solution, Stage 2 — Flash BIOS via the iDRAC Web UI

Now that iDRAC is 2.86, its web UI uses modern TLS and loads in a normal browser. The web UI's firmware-upload page does **not** require Enterprise — this is the door that was locked on 2.02.

1. Browse to `https://<IDRAC_IP>` and log in (`root` / `calvin` by default).

2. Go to **iDRAC Settings** → **Update and Rollback**.
3. Click **Browse / Choose File** and select the BIOS package, e.g. BIOS_KM6P8_WN64_2.19.0.EXE.
4. Click **Upload**. You should see “**Package successfully downloaded**” with **Criticality: Urgent**.
5. Click **Install and Reboot**. A job is queued (e.g. RAC0603 / JID_..., *Firmware Update: BIOS*).

7.1 If the job sits in “New” / won’t start

You may see:

```
RED023: Lifecycle Controller in use. This job will start when
Lifecycle Controller is available.
```

This happens if the LC home screen is open on the console. **Fix:** on the R730’s monitor, click **Exit** on the Lifecycle Controller home screen. The queued BIOS job then runs automatically: the server reboots into LC, shows “**Lifecycle Controller: System Configuration requested**”, and displays a progress bar (≈12 minutes). **Do not interrupt it.** Watch the queue with:

```
ssh root@<IDRAC_IP> "racadm jobqueue view"
```

7.2 Confirmation of success

The Lifecycle Log will show:

```
PR36   : Version change detected for BIOS firmware.
        Previous version:1.0.4, Current version:2.19.0
SUP0518: Successfully updated the Dell Server PowerEdge BIOS
        R630/R730/R730XD Version 2.19.0 ...
```

and the splash screen now reads **BIOS Version: 2.19.0**.

8 Reinstall the RAM

With BIOS 2.19.0 live, power the server **off**, reinstall **all 16 × 32 GB LRDIMMs** across both CPU banks/all slots, and power on. The updated BIOS trains the high-density LRDIMMs and the server POSTs to the full memory total.

If it still cycles

If the machine cycles through “Collecting System Inventory” and powers off *while only a minimal/partial DIMM set is installed*, that is a side effect of the partial config — populate the intended configuration and re-test. If it still misbehaves with the full, correct configuration, pull the logs:

```
ssh root@<IDRAC_IP> "racadm getsel"
ssh root@<IDRAC_IP> "racadm lclog view -c LCLLOG -n 30"
```

9 The Second Trap — CPU Stepping Mismatch

Even after the BIOS is successfully updated, a dual-socket R730 will hang indefinitely at:

```
Initializing Intel QuickPath Interconnect...
```

if the two installed CPUs have **different steppings** (different S-spec codes).

Symptoms — all present simultaneously

- Hangs at QPI init screen indefinitely (25+ minutes, no progress)
- `racadm get BIOS.MemSettings` shows `SysMemSize=32 GB` regardless of DIMM count
- iDRAC reports both CPUs as `Presence_Detected / Ok`
- **No fault logged anywhere** — SEL and Lifecycle Log show only informational entries
- Hang identical regardless of DIMM count, slots, CPU bank, AC drain, or NVRAM state
- Swapping the two CPUs between sockets produces identical hang

The fix

Read the S-spec code **physically off the processor lid** before installation. The code is printed on the top of the chip (e.g. SR2JS). Both processors must show **identical S-spec codes**. eBay listings for “matched pair E5-2699 v4” frequently mix steppings — verify both codes before purchase.

10 Final Working Configuration

Component	Detail
BIOS	2.19.0
iDRAC	2.86.86.86
Lifecycle Controller	2.86.86.86
CPUs	Dual Intel Xeon E5-2699 v4 (matched steppings) — 88 threads
RAM	512 GB DDR4-2133 ECC LRDIMM (16 × 32 GB Samsung M386A4G40DM0-CPB)
GPU	NVIDIA Quadro RTX 6000 24 GB
OS	Proxmox VE
Kernel	6.14.11-9-pve (pinned)
NVIDIA Driver	550.163.01
CUDA	12.4

Kernel pinning note

NVIDIA 550 fails against Proxmox kernels 6.17 and 7.0 (DRM API changes). Pin to 6.14:

```
apt install -y proxmox-kernel-6.14 proxmox-headers-6.14.11-9-pve
apt install -y nvidia-driver firmware-nvidia-graphics nvidia-persistenced
# In /etc/default/grub set GRUB_DEFAULT to 6.14.11-9-pve entry, then update-grub
```

11 What You Should See at Each Stage

Stage	On the console	racadm getversion
Start	No useable DIMMs found; loops	BIOS 1.0.4 / iDRAC 2.02 / LC 2.02
After Stage 1	Unchanged splash, still no POST	BIOS 1.0.4 / iDRAC 2.86 / LC 2.86
During Stage 2	LC progress bar, ~12 min	—
After Stage 2	Splash shows BIOS 2.19.0	BIOS 2.19.0 / iDRAC 2.86 / LC 2.86
Mismatched CPUs	Hangs at QPI indefinitely	—
Matched CPUs + RAM	Normal POST, full memory	—

12 Quick Reference / Command Cheat-Sheet

```
# --- Confirm iDRAC alive + versions ---
ssh root@<IDRAC_IP> "racadm getversion"

# --- Extract iDRAC .d7 out of the Linux .BIN ---
sudo /path/iDRAC-..._LN64_2.86.86.86_A00.BIN --extract /tmp/idrac_extract
find /tmp/idrac_extract -name 'firmimg.d7'

# --- TFTP server (Debian: serves /srv/tftp, may be masked) ---
sudo apt-get install -y tftpd-hpa
sudo systemctl unmask tftpd-hpa
sudo systemctl start tftpd-hpa

# --- Match what iDRAC requests: idrac.bin/firmimg.d7 ---
sudo mkdir -p /srv/tftp/idrac.bin
sudo cp /tmp/idrac_extract/payload/firmimg.d7 /srv/tftp/idrac.bin/firmimg.d7

# --- Open UDP/69 to the iDRAC (iptables) ---
sudo iptables -I INPUT -p udp --dport 69 -s <IDRAC_IP> -j ACCEPT
sudo iptables -I INPUT -p udp --sport 69 -s <IDRAC_IP> -j ACCEPT

# --- (Optional) verify TFTP serves the file ---
tftp <HELPER_HOST_IP> -c get idrac.bin/firmimg.d7 /tmp/test.d7 && ls -la /tmp/test.d7
```

```
# --- Flash iDRAC/LC 2.02 -> 2.86 ---
ssh root@<IDRAC_IP> "racadm fwupdate -g -u -a <HELPER_HOST_IP> -d idrac.bin"

# --- Wait ~5 min, then confirm ---
ssh root@<IDRAC_IP> "racadm getversion"      # iDRAC/LC now 2.86

# --- BIOS flash is done in the iDRAC 2.86 WEB UI: ---
#   https://<IDRAC_IP> -> iDRAC Settings -> Update and Rollback
#   Upload BIOS_..._2.19.0.EXE -> Install and Reboot
#   If the job stalls: click Exit on the LC home screen on the console.

# --- Watch the BIOS job ---
ssh root@<IDRAC_IP> "racadm jobqueue view"
```

13 Lessons Learned

- **Check CPU steppings before anything else.** Both CPUs in a dual-socket R730 must have identical S-spec codes. Read the lid. A mismatch produces no error, no logged fault — just a silent QPI hang that survives every other diagnostic step.
- **The iDRAC is your lifeline.** It runs on standby power independent of POST. If you can reach it on the network, the server is *not* bricked.
- **Order matters: iDRAC/LC first, BIOS second.** Updating the management stack is what re-opens a usable update path.
- `fwupdate` $\neq$ `update`. Legacy `fwupdate` (TFTP) handles iDRAC .d7 images and ignores the Enterprise-license gate. Use the right tool for the artifact.
- `idrac.bin` is requested as a **directory**. The firmware must live at `/srv/tftp/idrac.bin/firmimg.d7`. A “host not reachable” here is usually a *path* problem — confirm with `tcpdump -n udp port 69`.
- **Don’t chase phantom hardware faults.** Memory undetected in *every* permutation with **no amber LEDs** is a firmware signature, not eight dead DIMMs.
- **Third-party trial Enterprise licenses are device-locked.** They LIC017 on a tag they weren’t issued for — and the web-UI upload path doesn’t need Enterprise anyway.
- **Change the default credentials.** Everything here used `root/calvin`. Rotate it once stable.

14 Appendix A — Error-Code Glossary

Code / message	Where	Meaning
No useable DIMMs found	POST screen	BIOS can’t initialize installed memory.
No memory detected	iDRAC SEL	Same condition, logged out-of-band.
SWC0242	<code>racadm update</code>	Required (Enterprise) license missing.
HWC0005	LC GUI update	LC internal error; LC 2.02 update bug. (2.02)

Code / message	Where	Meaning
LIC017	License import	License corrupted/invalid or device-locked to another tag.
RED023	Job queue	“Lifecycle Controller in use” — release LC to let the job run.
RAC0603	iDRAC web UI	BIOS update job successfully queued.
PR36	Lifecycle Log	BIOS version change detected (success marker).
SUP0518	Lifecycle Log	BIOS update applied successfully (success marker).
SWC0053 / SWC0066	Catalog update	Catalog/repository problems.
SUP0530 / SUP0539	Catalog update	R730 no longer in Dell’s current catalog.
Specified path is too long	racadm fwupdate	Filename too long — use <code>idrac.bin</code> .
Remote host is not reachable	racadm fwupdate	Often a <i>path/firewall</i> issue, not a true outage.

15 Appendix B — Why Each Layer Blocks You

- **BIOS 1.0.4** — earliest memory-init phase can’t train the DIMMs, so the machine never reaches a menu, boot device, or recovery tool.
- **iDRAC 2.02 TLS** — serves only TLS 1.0/1.1; modern browsers refuse outright.
- **Basic/Express license** — gates `racadm update` and modern remote-update flows behind Enterprise.
- **LC 2.02 bug** — local/network LC updates fail with HWC0005.
- **CPU stepping mismatch** — no error logged; BIOS hangs silently at QPI bring-up. iDRAC reports both CPUs present. Only fix is a matched pair.
- **The way out** — `fwupdate` TFTP path predates all license checks, accepts the iDRAC `.d7`, updates LC/iDRAC to 2.86; the upgraded web UI flashes the BIOS; matched CPUs complete the POST.

*Document status: **confirmed fully operational**. QuarkyLab running Proxmox VE, dual E5-2699 v4 matched steppings (88 threads), 512 GB ECC RAM, NVIDIA Quadro RTX 6000, CUDA 12.4, kernel 6.14.11-9-pve, NVIDIA driver 550.163.01.*